Poor

P1: 0/4

P3(a,d): 1/12

Imperial College London

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

final : 2 out of 32

Machine Learning Assignment 3

Author:
Jorn Voegtli (CID: 00958215)

Date: March 10, 2017

Introduction

Pledge

I, Jorn Voegtli, pledge that this assignment is completely my own work, and that I did not take, borrow or steal work from any other person, and that I did not allow any other person to use, have, borrow or steal portions of my work. I understand that if I violate this honesty pledge, I am subject to disciplinary action pursuant to the appropriate sections of Imperial College London.
Collaborations: Anson Tang, Andrew Zhou, Javier Hernandez, Shiva Jagannaath

1 Problem 1

Incomplete

2 Problem 2

since the equation $\frac{1}{N}\sum_{i=1}^{N}(y - y_{(i)})^2$ is effectively the sum of the distance from each of the points. On a single line, if we move our point m from the lowest upwards we will continue getting a better error. This is until we reach the median. This is because as we move it away from the median (to higher values) although we are getting closer to the higher values we are getting away from more lower values and therefore increasing the overall error. This applies for both directions, and the median is therefore the value at which you get the minimum error.

3 Problem 3

3.1 Solution:

a)

The simplest rating predictor for movies and users (with the other kept constant) would be to take the average of their ratings from the training data. To calculate the error, we would simply compare that against the ratings in the test data to calculate the error.
My chosen method of calculating error was the Mean Square Error (MSE), after calculating the error for each user and movie I took the average for the overall user and movie error.:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_{test}^{(i)} - y)^2$$

Using this method. I got an average user error of 0.9601 and an average movie error of 0.6223. One would prefer the rating based on constant movies, as if you average

the ratings for a movie, you're likely to get a good estimation for what other people will think of the movie. However, for the constant user rating, as although there will be different opinions, the average would tend to a similar area and therefore the error would be smaller, it depends on what movie the user has watched and what genre he likes. Therefore he might decided to watch a movie in a genre he doesn't often watch and decide a movie is bad, which would make our predictor wrong.

b)

For my regression baseline I used Ridge Regression to preform the learning of the ratings of the user data. I used Ridge Regression as the inclusion of the λ prevents the regression from blowing up as $Z^T Z$ is nearly singular. I found the training error for each user, while using cross-validation to find the best λ for each user (and hence giving me the best weights and outcome), and then took the average of the total to get the overall training error. For the Test error, I used the optimal weights of each user to compare against the ratings. Again taking the Mean Square Error and computing the average of all the users to get the average error.
My final training and test errors were 0.6 and 0.9, respectively.

For my cross validation I decided to use the estimate for the Leave One Out Cross Validation method, given below:

$$CV = \frac{1}{N} \sum_{I=1}^{N} [\frac{\hat{y}_i - y_i}{1 - h_i}]^2$$

c)

As a transformation I decided to use the Legendre Polynomials. After implementing the Legendre Polynomials I found that they only very slightly improved the error (by about 0.1%), which is very insignificant to the overall error.

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Machine Learning Assignment 3

Author:
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P2 : 0

P3 : (b) 1

(c) 0

Date: March 10, 2017

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Incomplete

This is not enough 0

2 Problem 2

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